FOR COMPLETE BEGINNERS

Name it once.

Complete Modern Java, episode 13: methods, in Java and TypeScript

> Give instructions a name, pass values in, get a result back.

The same line printed in two places is two places to change.

A method names it once. Three shapes: no input, input, and a result.

In this episode

- void methods
- Calling a method
- Parameters and arguments
- return
- TypeScript functions

Coming next

- Arrays (episode 14)
- Array iteration (episode 15)
- Classes and objects (episode 16)

At the time of recording: Java 25, Node 24 LTS, IntelliJ IDEA 2026.1.3.

01 What each file printed

Lines.java: printLine() twice	`----------, Sarah's groceries, ----------`
Custom.java: two arguments	`==========, **********`
Totals.java: lineTotal(2, 150)	`300`
breadTotal + milkTotal	`399`
receipt.ts	`----------, ==========, 399`

Every row was run, not predicted.

02 Three shapes

void: nothing given back. A parameter is the name inside; an argument is the value in the call.

return sends a value back; the type before the name says what kind.

Totals.java

```
void main() {
    int breadTotal = lineTotal(2, 150);
    int milkTotal = lineTotal(1, 99);
    IO.println(breadTotal + milkTotal);   // 399
}

int lineTotal(int quantity, int priceCents) {
    return quantity * priceCents;
}
```

> Printed 300, then 399.
>
> Run on a Mac, 2026-10-24: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

03 TypeScript functions

The return type goes after the brackets, where Java puts it before the name.

receipt.ts

```
function lineTotal(quantity: number, priceCents: number): number {
    return quantity * priceCents;
}
```

> npx tsc --noEmit was quiet; node printed ----------, ==========, 399.
>
> Run on a Mac, 2026-10-24: Java 25.0.4.1, Node 24.22.1, TypeScript 7.0.2.

Write a method

1. Repeating lines? Name them once.
2. Needs a value? Add a parameter.
3. Needs to give an answer? return it, and say its type.
4. Call it from main, then carry on.
5. Challenge: lineTotal(3, 150) + lineTotal(2, 99).

Everything else

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The code and the challenge files: complete-modern-java
https://github.com/code-with-sam-dev/complete-modern-java

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